Toxicology foundations

Orientation and immediate approach

- Physiology applied under uncertainty
  - Histories incomplete, products mixed
- Protect staff and stop exposure
  - Remove clothing, brush off dry chemicals
- Stabilise airway, breathing, circulation
- Normal early findings do not exclude delay
  - Delayed release, metabolites, hepatic toxicity
- History: substance, maximum amount, time, route
- Self-poisoning: psychosocial review after stabilisation
  - Low intent does not reduce dose

Testing and decontamination

- Paracetamol measured despite no symptoms
- QRS widening: sodium-channel blockade
- QT prolongation: polymorphic ventricular tachycardia
- Urine screens: treat the patient, not the screen
- Charcoal: early, cooperative, protected airway
  - Aspiration can be catastrophic
  - Poor binding of alcohols, metals, corrosives
- Whole-bowel irrigation for selected cases

Opioids, sedatives, antidepressants

- Ventilation is the priority
- Naloxone titrated to breathing
  - May wear off before the opioid
- Flumazenil can precipitate seizure
- Tricyclics block fast sodium channels
  - Bicarbonate treats cardiovascular toxicity
- Serotonin reuptake inhibitors often milder

Stimulants and hyperthermia

- Catecholamine excess: agitation, vasospasm, heat
- Benzodiazepines reduce central activation
- Rapid external cooling, not antipyretics
- Serotonin toxicity: stop, sedate, cool
- Neuroleptic malignant syndrome: stop, support
- Malignant hyperthermia: immediate dantrolene

Envenomation and bites

- Record time, place, progression
- Keep still, immobilise, pressure guidance
  - No cutting, sucking, freezing, tourniquet
- Antivenom by syndrome, ready for anaphylaxis
- Bites: irrigate, tetanus, rabies, antibiotics
  - Hand and fist injuries: urgent surgery

Toxidromes

- Opioid: low consciousness, slow breathing, small pupils
- Sedative-hypnotic: no unique pupil pattern
- Sympathomimetic: agitation, mydriasis, sweating
- Antimuscarinic: mydriasis with dry flushed skin
  - Reduced bowel sounds, urinary retention
- Cholinergic: secretions, bronchospasm, small pupils
- Serotonin toxicity: hyperreflexia, clonus, heat
- Neuroleptic malignant syndrome: slower, rigid
- Malignant hyperthermia: anaesthetic-triggered

Paracetamol and salicylate

- Paracetamol saturates conjugation
  - Reactive metabolite depletes glutathione
- Nomogram only for single timed ingestion
- N-acetylcysteine replenishes glutathione
- Salicylate uncouples and stimulates respiration
  - Early respiratory alkalosis, later acidosis
- Toxicity can worsen as serum level falls
- Avoid intubation, preserve hyperventilation

Alcohols and gases

- Ethanol intoxication is a diagnosis of exclusion
- Methanol: formic acid, vision, basal ganglia
- Ethylene glycol: acidosis, kidney, oxalate
- Fomepizole blocks alcohol dehydrogenase
- Carbon monoxide: pulse oximetry can look normal
  - High oxygen, co-oximetry, hyperbaric advice
- Cyanide blocks electron transport
  - Lactic acidosis, prompt antidote

Pesticides, metals, corrosives

- Organophosphates inhibit acetylcholinesterase
  - Atropine to secretions and perfusion
  - Weakness after apparent recovery
- Lead: children and pregnancy vulnerable
  - Remove source, chelate by level and symptoms
- Acids coagulate, alkalis penetrate
- Oropharynx does not predict oesophagus

Heat, cold, immersion

- Heat exhaustion lacks neurological dysfunction
- Heat stroke: encephalopathy, organ injury
  - Cold-water immersion, do not delay
- Hypothermia: bradycardia, coagulopathy
  - Handle gently, rewarm by severity
- Drowning: ventilation and oxygenation first